CSE370 : Database Systems
Project Report
Project Title : TrippyBD

Group No : 09, CSE370 Lab Section : 05 , Spring 2026		
ID	**Name**	**Contribution**
23301077	Saif Ibnay Sahed	Backend development (Django), frontend (HTML, Bootstrap), database implementation (MySQL), core functionality development
23301043	Fatima Jannat	Django model design, GitHub management, frontend (HTML, Bootstrap); collaborated in debugging and feature development.

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Introduction

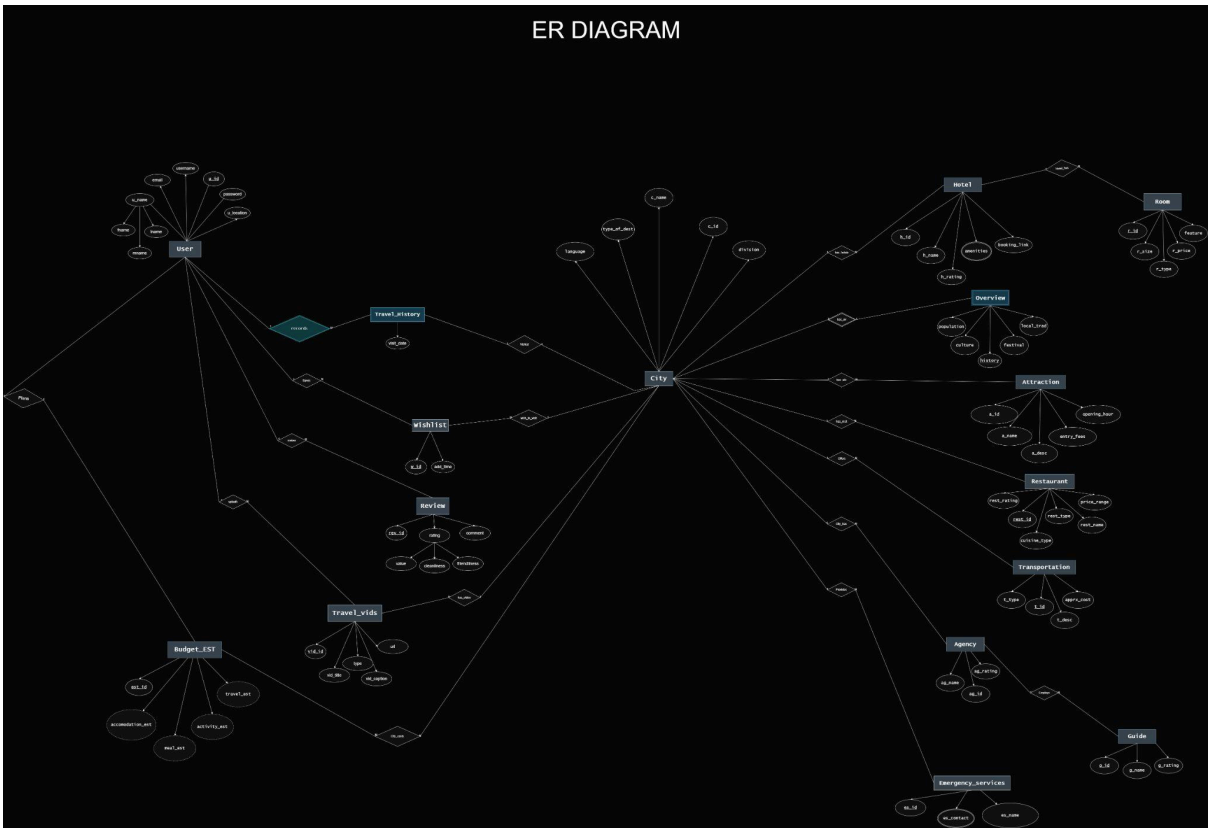
Trippy BD is a new web based travel platform which aims to make travel planning within Bangladesh an easier and better experience. The platform allows users to easily browse through different cities, get detailed information about local attractions, hotels and restaurants, and plan their trips with ease. Trippy BD combines various travel-related services in one platform, providing an all-in-one, easy-to-use and organized solution, enabling travelers to make informed choices and have a smooth travel-planning experience.

Project Features

ID, Name	Features [3 per member]	
23301077, Saif Ibnay Sahed	Ft 1	City Overview
	Ft 2	Plan Maker
	Ft 3	Review
	Ft 4	Hotel Booking
	Ft 5	Tour Guide Hiring
	Ft 6	Travel History
	Ft 7	Accounts Dashboard
	Ft 8	User Authentication and Password Protection
23301043, Fatima Jannat	Ft 1	Emergency Services
	Ft 2	Wishlist
	Ft 3	Attraction & Restaurant Listing & Details
	Ft 4	Restaurant Listing & Details
	Ft 5	Transportation Listing & Details

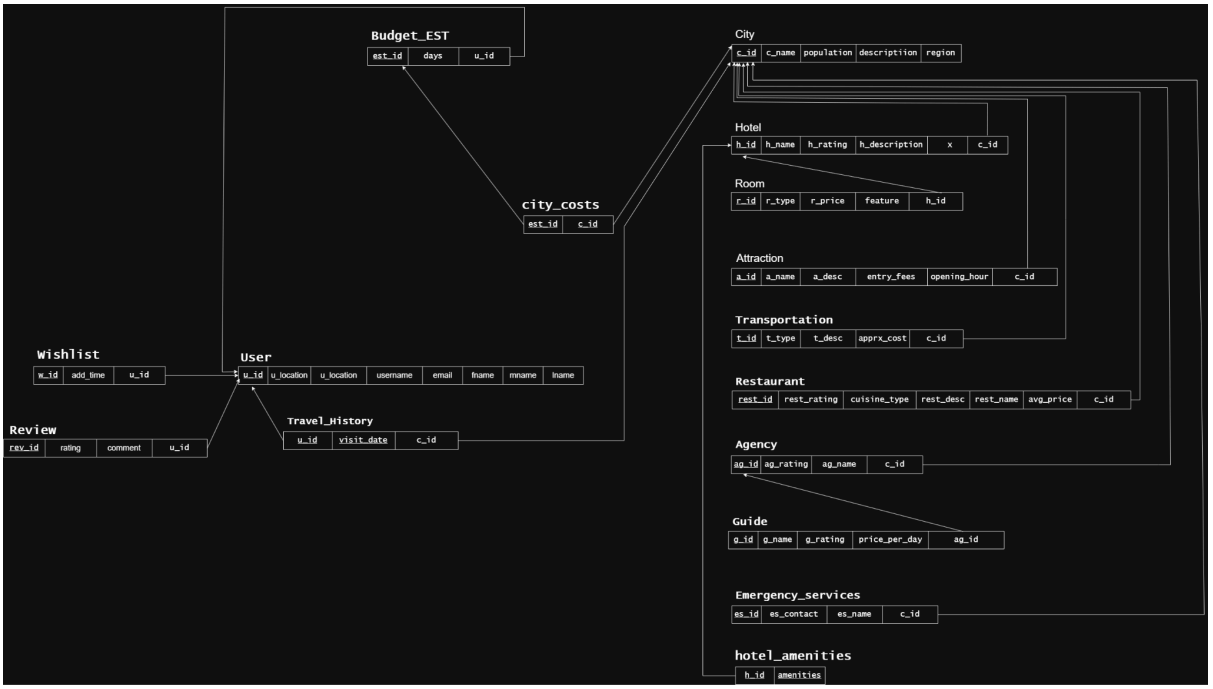
ER/EER Diagram

ER Diagram Link for better view:
https://drive.google.com/file/d/14Afix9NbIS7tPk3Si7zEm2i0YCZsqkfn/view?usp=sharing



Schema Diagram

Schema Diagram Link for better view:
https://drive.google.com/file/d/1LBuFh0GkWgBxZBO2QoDRqrDZSNZra0Fw/view?usp=sharing



Normalization

a. Explain if your converted Schema is in 1NF or not. If not, decompose it to 1NF.
   Ans: As there are no composite, multivalued and nested relations - therefore the schema is already in 1NF form.

b. Explain if your converted Schema is in 2NF or not. If not, decompose it to 2NF. Can there be any partial functional dependencies in your relational schema?
   Ans: The schema is in 2NF as it satisfies 1NF and contains no partial functional dependencies.

c. Explain if your converted Schema is in 3NF or not. If not, decompose it to 3NF. Can there be any transitive dependencies in your relational schema?
   Ans: The schema is in 3NF because it satisfies 2NF and has no transitive dependencies. No non-key attribute depends on another non-key attribute to be identified; for instance, in the Hotel, City, and Restaurant tables, every descriptive field depends strictly and directly on the primary key.

Frontend Development

The frontend of Trippy BD was developed using HTML, CSS and JavaScript, and Bootstrap was integrated to provide a responsive and visually appealing user interface. The primary aim of the frontend was to create a straightforward, user-friendly, and accessible platform where users could effortlessly browse travel-related information without complications.

Bootstrap templates were used as the basis for these sites to speed up development and keep the sites looking modern. They were customized with CSS to match the theme of the project and improve overall user experience. Interactivity was added using JavaScript, for example for dynamic content updates, form handling, filtering options and improving navigation flow.

The interface is designed to be responsive and adapt to various screen sizes such as desktops, tablets, and mobile devices. Important pages such as the homepage, city exploration pages, hotel listings and dashboard were designed for intuitive navigation and fast access to key features.

```html
<div class="collapse navbar-collapse" id="mainNav">
  <ul class="navbar-nav me-auto ms-3 gap-1">
    <li class="nav-item">
      <a class="nav-link-custom" href="/cities/">🌆 Cities</a>
    </li>
    <li class="nav-item">
      <a class="nav-link-custom" href="/hotels/">🏨 Hotels</a>
    </li>
    <li class="nav-item">
      <a class="nav-link-custom" href="/agencies/">🏢 Agencies</a>
    </li>
    <li class="nav-item">
      <a class="nav-link-custom" href="/restaurants/">🍽 Restaurants</a>
    </li>
    <li class="nav-item">
      <a class="nav-link-custom" href="/attractions/">🎡 Attractions</a>
    </li>
    <li class="nav-item">
      <a class="nav-link-custom" href="{% url 'budget_planner' %}">📊 Plan Maker</a>
    </li>
  </ul>
```

```style
<style>
  * {
    font-family: 'Poppins', sans-serif;
    margin: 0;
    padding: 0;
    box-sizing: border-box;
  }
  body {
    background-color: #0f172a;
    color: #e2e8f0;
    min-height: 100vh;
    display: flex;
    flex-direction: column;
  }

  /* — Navbar — */
  .navbar-custom {
    background: rgba(15, 23, 42, 0.98);
    backdrop-filter: blur(12px);
    border-bottom: 1px solid rgba(255,255,255,0.1);
    padding: 16px 0;
  }
```

```html
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8" />
  <meta name="viewport" content="width=device-width, initial-scale=1.0"/>
  <title>{% block title %}TrIppyBD{% endblock %}</title>

  <!-- Bootstrap 5 -->
  <link rel="stylesheet"
    href="https://cdnjs.cloudflare.com/ajax/libs/bootstrap/5.3.2/css/bootstrap.min.css"/>

  <!-- Google Fonts -->
  <link href="https://fonts.googleapis.com/css2?family=Poppins:wght@300;400;500;600;700&display=swap" rel="stylesheet"/>
  <link rel="stylesheet" href="https://cdnjs.cloudflare.com/ajax/libs/font-awesome/6.0.0/css/all.min.css">
```

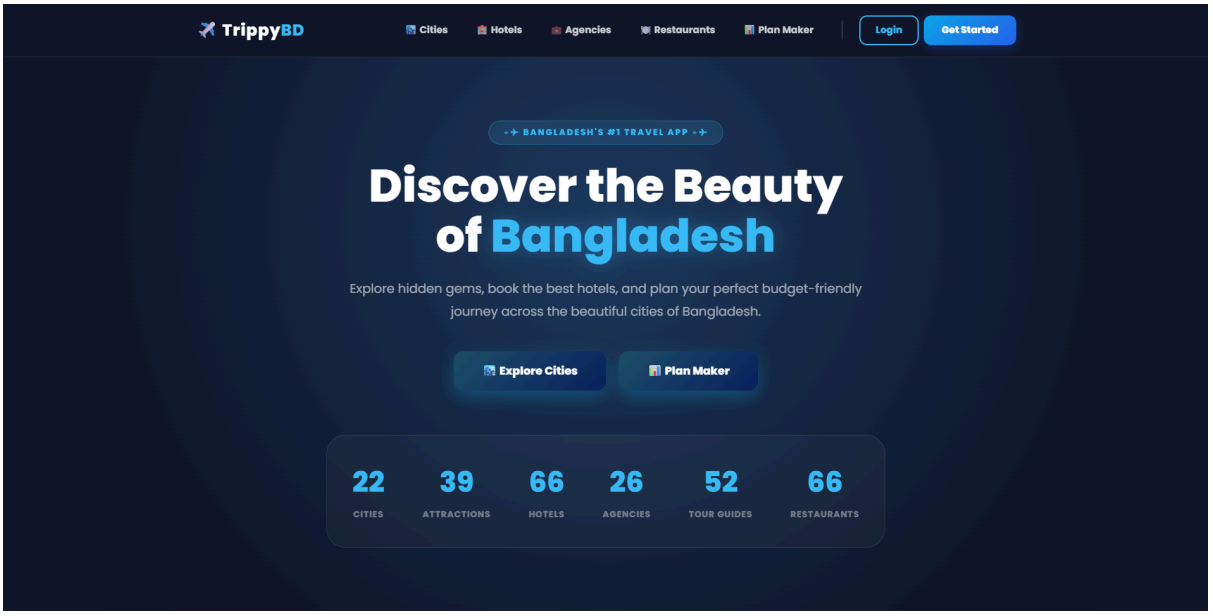
Contribution of ID : 23301043, Name : Fatima Jannat

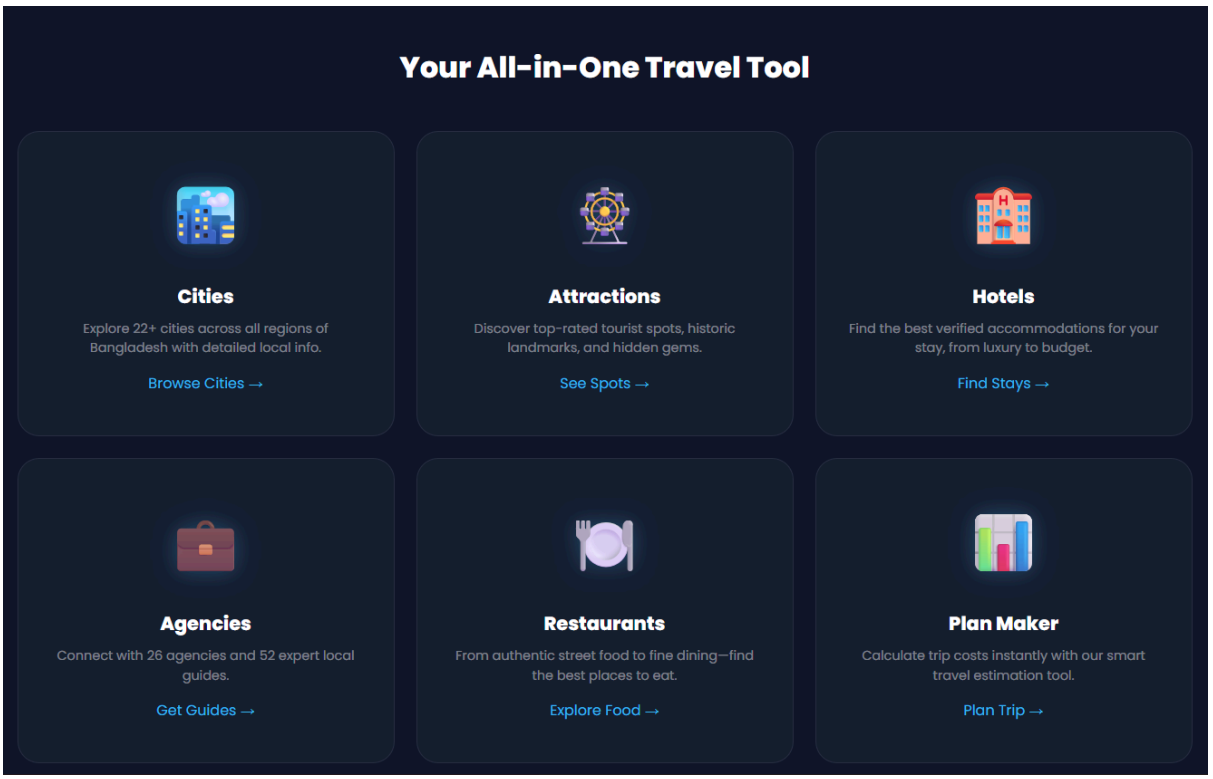
- Designed and structured frontend pages using HTML and CSS.
- Custom Bootstrap templates to suit project themes.
- Worked on user interface elements such as navigation bars, cards.
- Implemented responsive layouts for different devices.

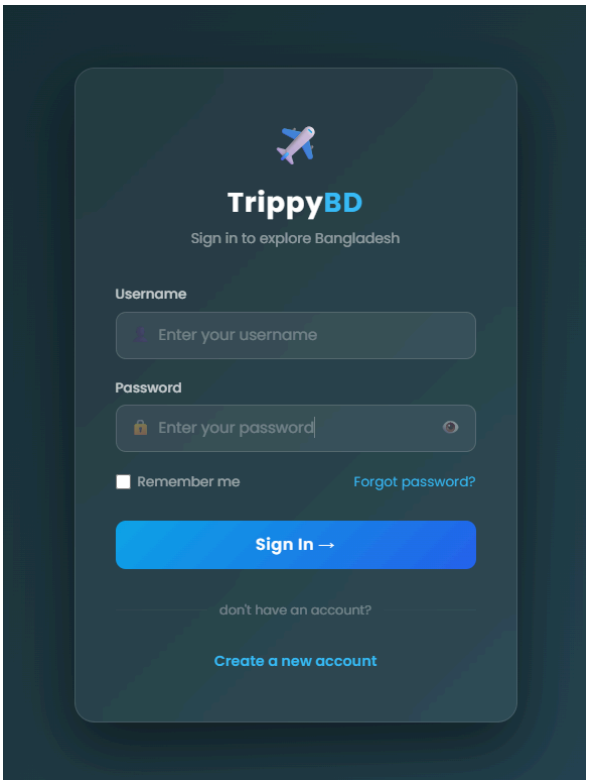
Contribution of ID : 23301077, Name : Saif Ibnay Sahed

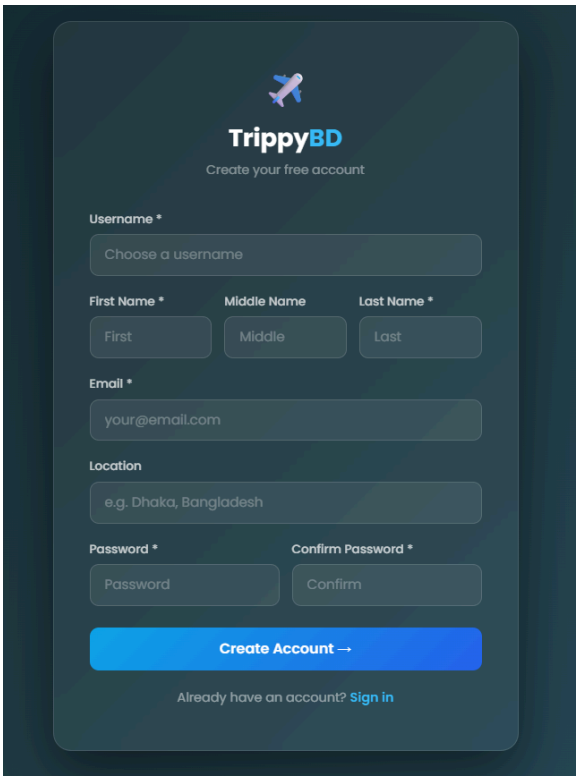
- Designed and structured frontend pages using HTML and CSS.
- Custom Bootstrap templates to suit project themes.
- Developed responsive layouts for different devices.
- Integrated JavaScript for interactivity and dynamic behavior.
- Worked on user interface elements such as navigation bars, cards and forms.
- Implemented responsive layouts for different devices.

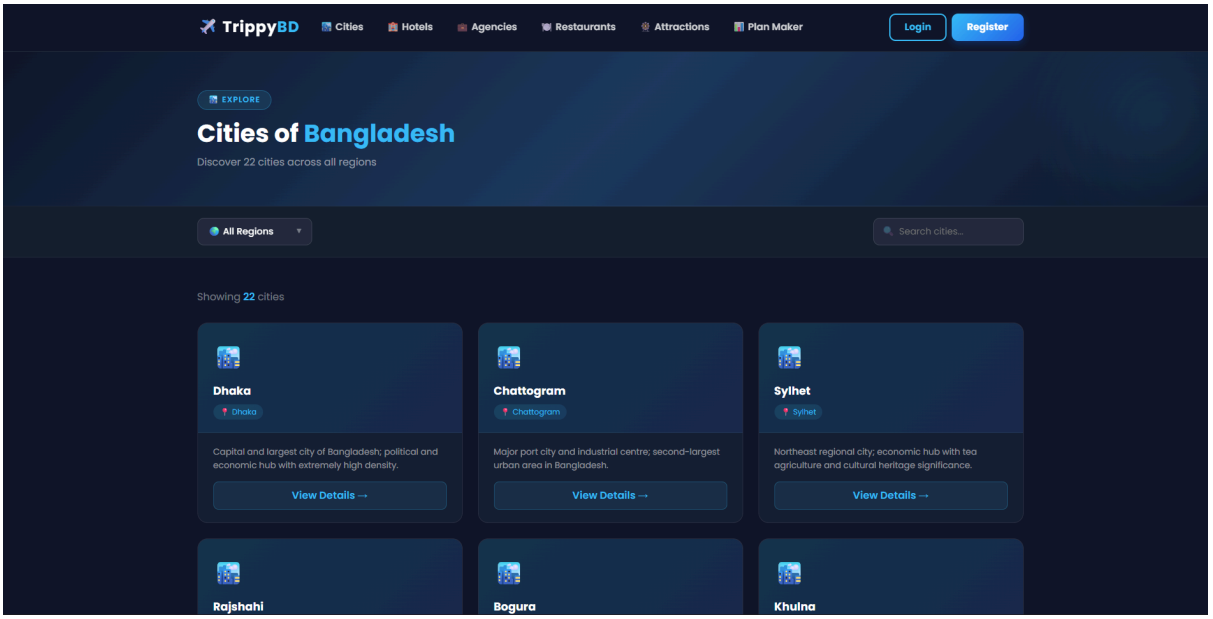
Some Reference Images:

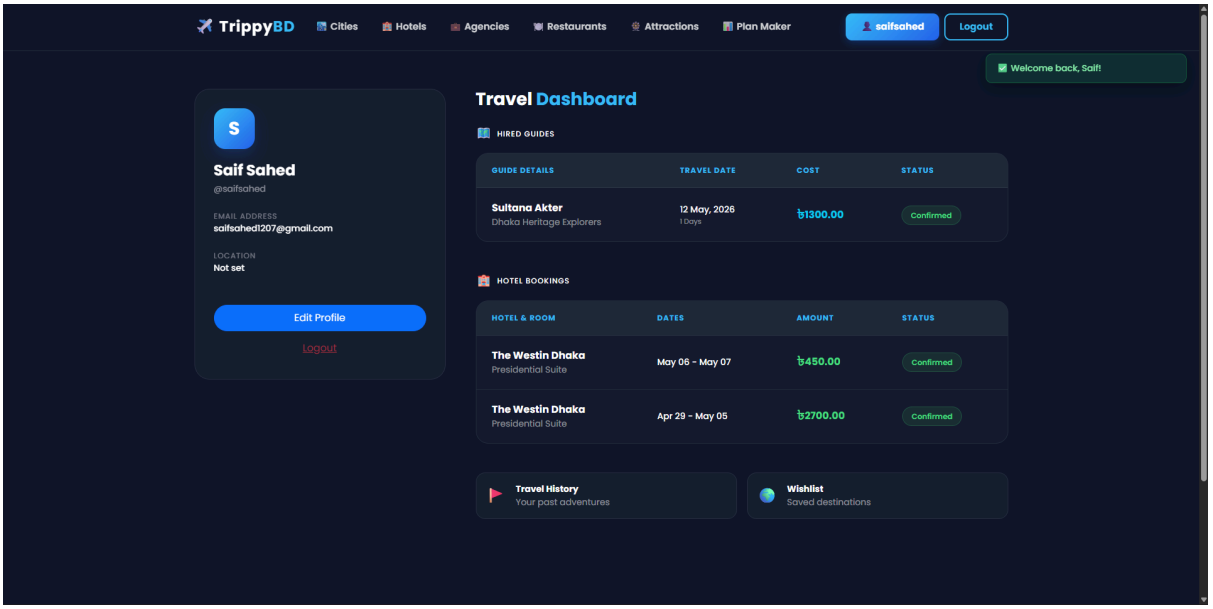


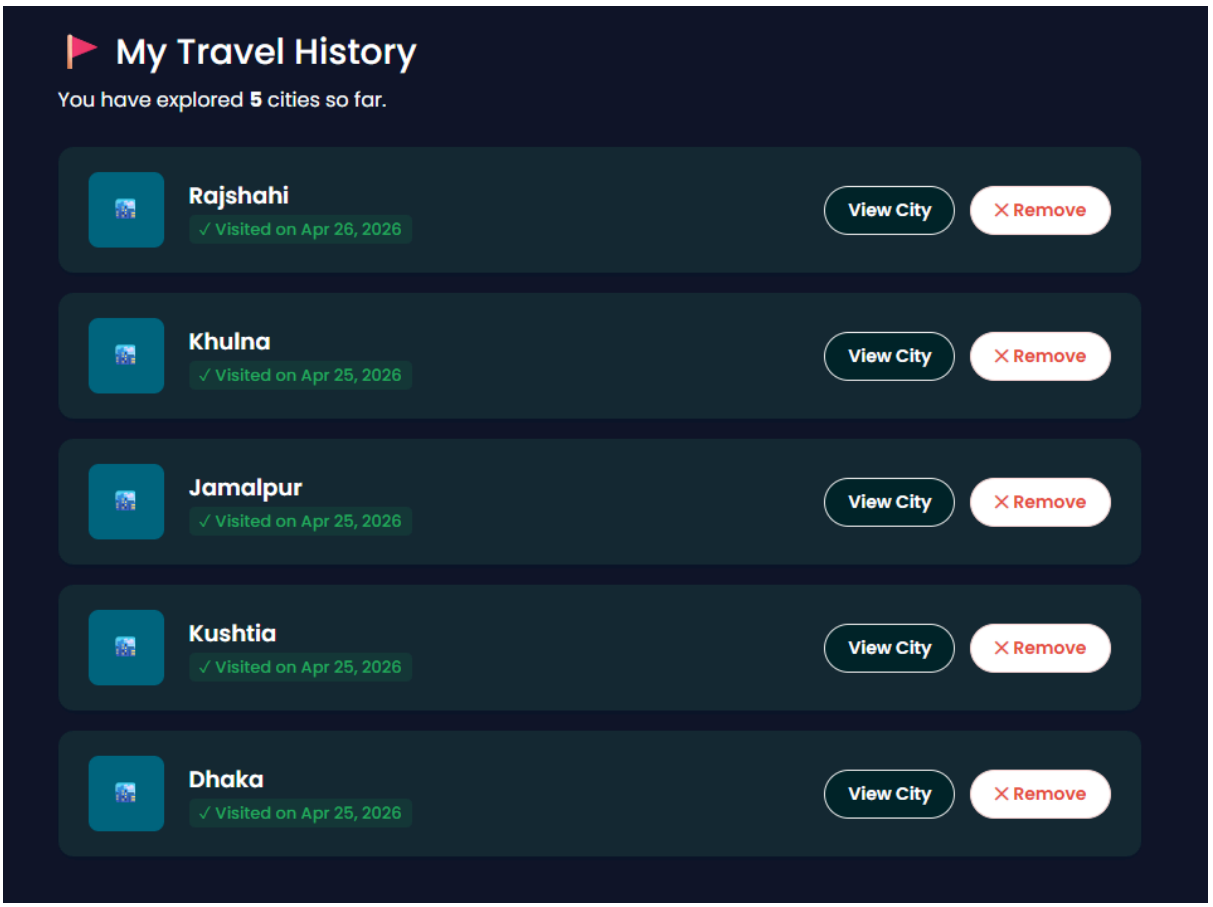

Your All-in-One Travel Tool
Cities
Explore 22+ cities across all regions of Bangladesh with detailed local info.
Browse Cities →
Attractions
Discover top-rated tourist spots, historic landmarks, and hidden gems.
See Spots →
Hotels
Find the best verified accommodations for your stay, from luxury to budget.
Find Stays →
Agencies
Connect with 26 agencies and 52 expert local guides.
Get Guides →
Restaurants
From authentic street food to fine dining—find the best places to eat.
Explore Food →
Plan Maker
Calculate trip costs instantly with our smart travel estimation tool.
Plan Trip →


TrippyBD
Sign in to explore Bangladesh
Username
Enter your username
Password
Enter your password
Remember me
Forgot password?
Sign In →
don't have an account?
Create a new account

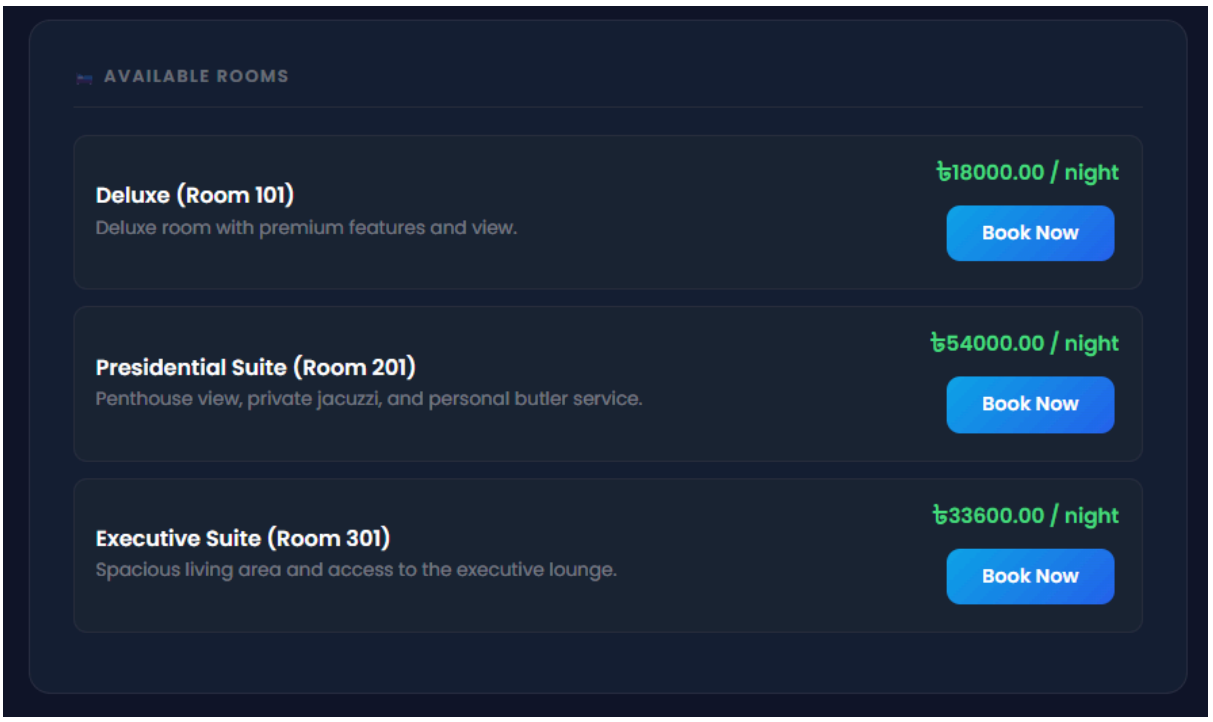

TrippyBD
Create your free account
Username *
Choose a username
First Name *
Middle Name
Last Name *
First
Middle
Last
Email *
your@email.com
Location
e.g. Dhaka, Bangladesh
Password *
Confirm Password *
Password
Confirm
Create Account →
Already have an account? Sign in

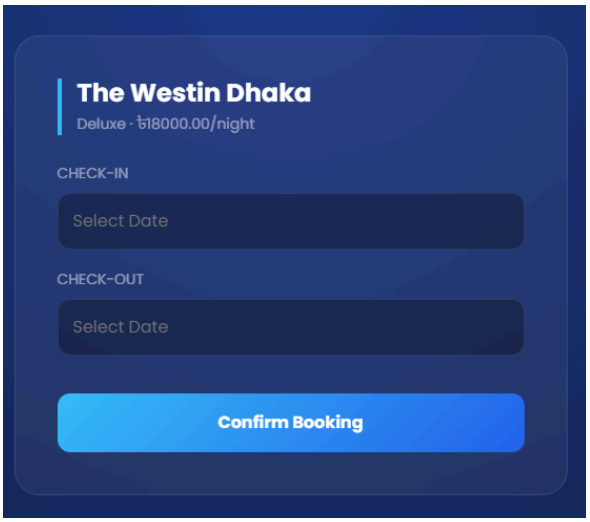

TrippyBD
Cities
Hotels
Agencies
Restaurants
Attractions
Plan Maker
Login
Register
EXPLORE
Cities of Bangladesh
Discover 22 cities across all regions
All Regions
Search cities...
Showing 22 cities
Dhaka
Dhaka
Capital and largest city of Bangladesh; political and economic hub with extremely high density.
View Details →
Chattogram
Chattogram
Major port city and industrial centre; second-largest urban area in Bangladesh.
View Details →
Sylhet
Sylhet
Northeast regional city; economic hub with tea agriculture and cultural heritage significance.
View Details →
Rajshahi
Bogura
Khulna

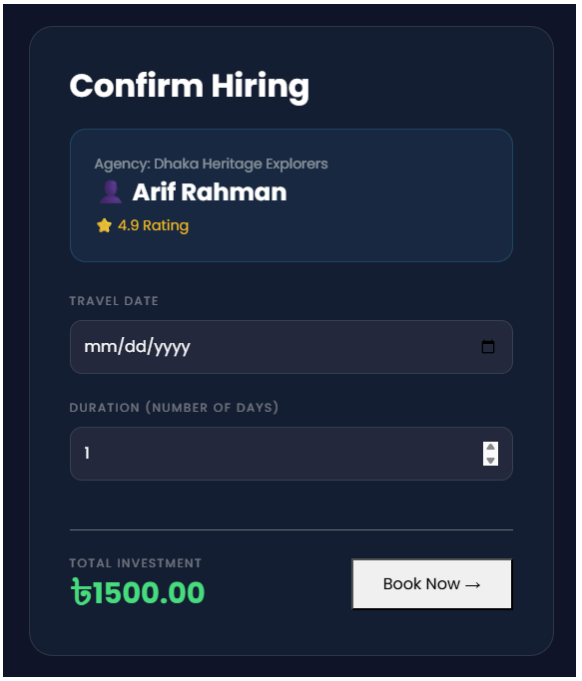

TrippyBD
Cities
Hotels
Agencies
Restaurants
Attractions
Plan Maker
saifsahed
Logout
Welcome back, Saif!
S
Saif Sahed
@saifsahed
EMAIL ADDRESS
saifsahed1207@gmail.com
LOCATION
Not set
Edit Profile
Logout
Travel Dashboard
HIRED GUIDES
GUIDE DETAILS
TRAVEL DATE
COST
STATUS
Sultana Akter
Dhaka Heritage Explorers
12 May, 2026
1 Days
৳1300.00
Confirmed
HOTEL BOOKINGS
HOTEL & ROOM
DATES
AMOUNT
STATUS
The Westin Dhaka
Presidential Suite
May 06 - May 07
৳450.00
Confirmed
The Westin Dhaka
Presidential Suite
Apr 29 - May 05
৳2700.00
Confirmed
Travel History
Your past adventures
Wishlist
Saved destinations

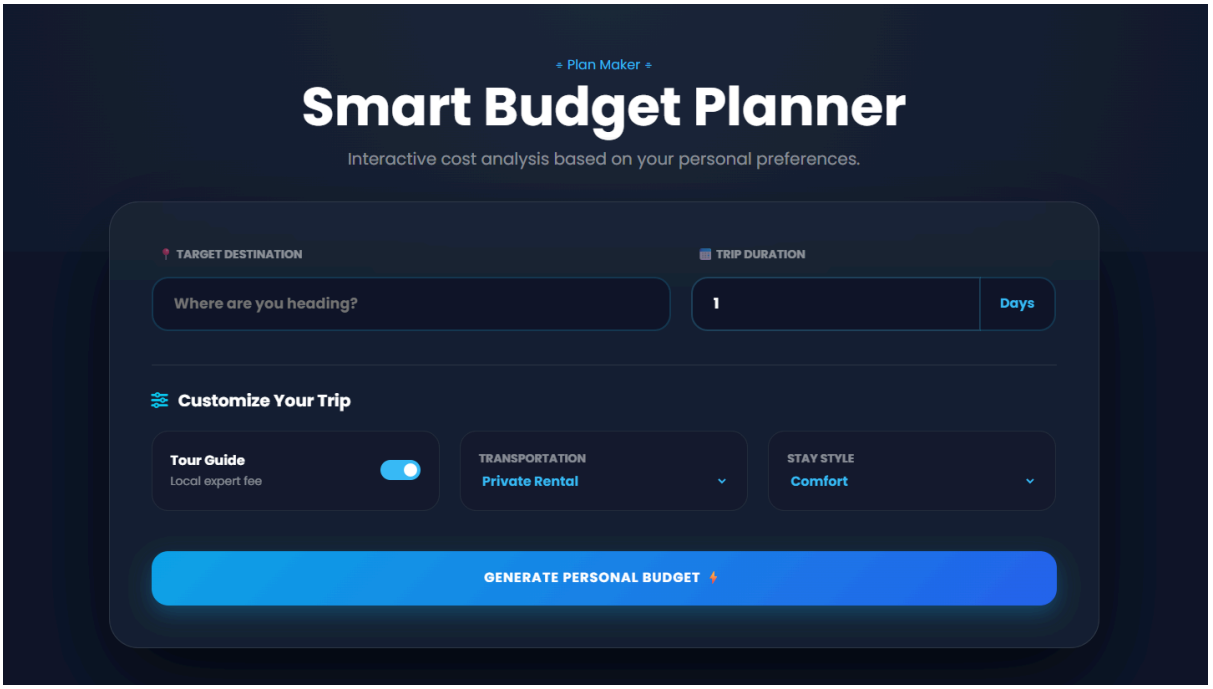

My Travel History
You have explored 5 cities so far.
Rajshahi
✓ Visited on Apr 26, 2026
View City
✕ Remove
Khulna
✓ Visited on Apr 25, 2026
View City
✕ Remove
Jamalpur
✓ Visited on Apr 25, 2026
View City
✕ Remove
Kushtia
✓ Visited on Apr 25, 2026
View City
✕ Remove
Dhaka
✓ Visited on Apr 25, 2026
View City
✕ Remove

My Travel Wishlist
Barisal — Barisal
Added 6 days, 13 hours ago
✓ Visited
✕ Drop
Sylhet — Sylhet
Added 1 week ago
✓ Visited
✕ Drop


AVAILABLE ROOMS
Deluxe (Room 101)
Deluxe room with premium features and view.
৳18000.00 / night
Book Now
Presidential Suite (Room 201)
Penthouse view, private jacuzzi, and personal butler service.
৳54000.00 / night
Book Now
Executive Suite (Room 301)
Spacious living area and access to the executive lounge.
৳33600.00 / night
Book Now


The Westin Dhaka
Deluxe · ৳18000.00/night
CHECK-IN
Select Date
CHECK-OUT
Select Date
Confirm Booking


Confirm Hiring
Agency: Dhaka Heritage Explorers
Arif Rahman
4.9 Rating
TRAVEL DATE
mm/dd/yyyy
DURATION (NUMBER OF DAYS)
1
TOTAL INVESTMENT
৳1500.00
Book Now →


Plan Maker
Smart Budget Planner
Interactive cost analysis based on your personal preferences.
TARGET DESTINATION
Where are you heading?
TRIP DURATION
1
Days
Customize Your Trip
Tour Guide
Local expert fee
TRANSPORTATION
Private Rental
STAY STYLE
Comfort
GENERATE PERSONAL BUDGET

Backend Development

For this project, the backend was built with Django, a high-level Python Web framework that promotes rapid development and clean pragmatic design. Django was picked because of its efficiency, built-in security, and ability to handle complex server-side logic.

The database was built locally on a MySQL server with the help of XAMPP. Next, we configured the appropriate database settings in Django's settings.py file to connect MySQL to Django, allowing for seamless data management and retrieval.

Key Elements

- **Django Models & Views:** Models were created to model and manage core data entities. Views were created to handle user requests and interact with the database as needed.

```python
from django.db import models

class Attraction(models.Model):
    city = models.ForeignKey('cities.City', on_delete=models.CASCADE)
    name = models.CharField(max_length=100)
    description = models.TextField()
    entry_fee = models.DecimalField(max_digits=6, decimal_places=2)
    opening_hours = models.CharField(max_length=100)

    def __str__(self):
        return self.name
```

```python
from django.shortcuts import render, get_object_or_404
from .models import Attraction
from cities.models import City
from django.contrib.contenttypes.models import ContentType
from review.models import Review


def attraction_list(request):
    attractions = Attraction.objects.select_related('city').all()
    cities      = City.objects.all()
    return render(request, 'attractions/attraction_list.html', {
        'attractions': attractions,
        'cities':      cities,
    })

def attraction_list_by_city(request, city_id):
    city        = get_object_or_404(City, pk=city_id)
    attractions = city.attraction_set.select_related('city').all()
    cities      = City.objects.all()
    return render(request, 'attractions/attraction_list.html', {
        'attractions':   attractions,
        'selected_city': city,
        'cities':        cities,
    })
```

- **Database Integration:** The reliable data storage and retrieval was provided by integrating MySQL database into Django project through standard configuration with XAMPP.

```
DATABASES = {
    'default': {
        'ENGINE': 'django.db.backends.mysql',
        'NAME': 'trippybd',
        'USER': 'root',
        'PASSWORD': '',
        'HOST': 'localhost',
        'PORT': '3306',
        'OPTIONS': {
            'sql_mode': 'STRICT_TRANS_TABLES',

        }
    }
}
```

```
MariaDB [trippybd]> show tables;
+-------------------------------------+
| Tables_in_trippybd                  |
+-------------------------------------+
| accounts_user                       |
| accounts_user_groups                |
| accounts_user_user_permissions      |
| agencies_agency                     |
| agencies_guide                      |
| agencies_guidehire                  |
| attractions_attraction              |
| auth_group                          |
| auth_group_permissions              |
| auth_permission                     |
| budget_budgetplan                   |
| cities_city                         |
| django_admin_log                    |
| django_content_type                 |
| django_migrations                   |
| django_session                      |
| emergency_services_emergencyservice |
| hotels_amenity                      |
| hotels_booking                      |
| hotels_hotel                        |
| hotels_hotel_amenities              |
| hotels_room                         |
| restaurants_restaurant              |
| review_review                       |
| transportation_transportation       |
| travel_history_travelhistory        |
| wishlist_wishlist                   |
+-------------------------------------+
27 rows in set (0.001 sec)
```

Contribution of ID : 23301043, Name : Fatima Jannat

- **Backend Logic:**
  - **Designed Models: Wishlist, Restaurant, Attractions, Emergency Services, Transportation**
  - **Created urls**

Contribution of ID : 23301077, Name : Saif Ibnay Sahed

- **Backend Logic:**
  - **Designed Models: City, Hotel, Accounts, Budget, Review, Agency, Travel History**
  - **Designed Views and Urls**
  - **Added User Authentication and Password Protection**

Some Reference Images:

```python
from django.db import models
from cities.models import City

class Restaurant(models.Model):
    city          = models.ForeignKey(City, on_delete=models.CASCADE)
    name          = models.CharField(max_length=200)
    description   = models.TextField()
    cuisine_type  = models.CharField(max_length=100)
    average_price = models.DecimalField(max_digits=8, decimal_places=2)
    rating        = models.DecimalField(max_digits=3, decimal_places=1)

    def __str__(self):
        return self.name
```

```python
from django.contrib import admin
from django.contrib.auth.admin import UserAdmin as BaseUserAdmin
from .models import User

class UserAdmin(BaseUserAdmin):
    list_display  = ('username', 'email', 'fname', 'lname', 'u_location', 'is_staff')
    search_fields = ('username', 'email', 'fname', 'lname')
    ordering      = ('username',)

    fieldsets = (
        (None,            {'fields': ('username', 'password')}),
        ('Personal Info', {'fields': ('fname', 'mname', 'lname', 'email', 'u_location')}),
        ('Permissions',   {'fields': ('is_active', 'is_staff', 'is_superuser', 'groups', 'user_permissions')}),
        ('Dates',         {'fields': ('date_joined',)}),
    )
    readonly_fields = ('date_joined', 'u_id')

admin.site.register(User, UserAdmin)
```

```python
from django.urls import path
from . import views

urlpatterns = [
    path('',                 views.city_list,    name='city_list'),
    path('<int:city_id>/', views.city_details, name='city_details'),
]
```

```python
INSTALLED_APPS = [
    'django.contrib.admin',
    'django.contrib.auth',
    'django.contrib.contenttypes',
    'django.contrib.sessions',
    'django.contrib.messages',
    'django.contrib.staticfiles',
    'cities',
    'attractions',
    'hotels',
    'restaurants',
    'accounts',
    'wishlist',
    'travel_history',
    'agencies',
    'transportation',
    'budget',
    'review',
    'emergency_services',
]
```

```python
urlpatterns = [
    path('admin/',        admin.site.urls),
    path('accounts/',     include('accounts.urls')),
    path('cities/',       include('cities.urls')),
    path('attractions/', include('attractions.urls')),
    path('hotels/',       include('hotels.urls')),
    path('restaurants/', include('restaurants.urls')),
    path('',              views.home, name='home'),
    path('wishlist/', include('wishlist.urls', namespace='wishlist')),
    path('cities/',       include(('cities.urls', 'cities'), namespace='cities')),
    path('travel-history/', include(('travel_history.urls', 'travel_history'), namespace='travel_history')),
    path('agencies/', include('agencies.urls')),
    path('transportation/', include('transportation.urls')),
    path('budget/', include('budget.urls')),
    path('reviews/', include('review.urls')),
    path('emergency/', include('emergency_services.urls')),

]
```

```python
AUTH_PASSWORD_VALIDATORS = [
    {
        'NAME': 'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.CommonPasswordValidator',
    },
    {
        'NAME': 'django.contrib.auth.password_validation.NumericPasswordValidator',
    },
]
```

Source Code Repository

The complete source code of the project has been uploaded to GitHub for version control, collaboration, and public access. The repository contains all relevant files, including frontend, backend, database configurations, and project documentation.

The project can be accessed using the following link:
GitHub Repository: TrippyBD

The repository is publicly accessible and organized in a structured manner to ensure ease of navigation and understanding.

Conclusion

Trippy BD demonstrates the practical application of full-stack web development to solve real-world problems in the travel domain. It has city exploration, personalized recommendations, budget planning, and essential travel resources integrated into a single platform.

This project reflects a strong understanding of system design, database management, and user-centered interface development. Through its modular structure and scalable features, Trippy BD has the potential to evolve into a fully functional, real-world travel solution.

Overall, the project successfully meets its academic objectives by combining technical implementation with practical relevance, laying a solid foundation for future improvements such as advanced recommendation systems, real-time data integration, and mobile responsiveness.

References

YouTube, Google, Coursera, and Udemy were utilized as learning resources, and all content used in this project was original and not plagiarized.